MIL-F-45421D (MU)
9 JULY 1973
SUPERSEDING
MIL-F-45421C (MU)
18 SEPTEMBER 1970

MILITARY SPECIFICATION

FIRING DEVICE, ELECTRICAL, M57 PARTS FOR, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the parts, assembling and packing for one type of firing device designated as Firing Device, Electrical, M57.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Boxes, Fiber

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-A-2550 - Ammunition and Special Weapons, General Specification for.
MIL-S-4473 - Shielding of Magnetron Tube and Magnets and Components Containing Magnetron Tubes or Magnets for Air Shipment.
MIL-L-10547- Liners, Case, and Sheet, Overwrapped; Watervaporproof
MIL-F-50449- Filler, Sheet Form (For Use in Ammunition Container)

FSC: 1345

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

8800924 - Firing Device, Anti-Personnel Mine, Electrical, M57
8800925 - Connector, Male
9200408 - Chamber, Vacuum Test

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material. - Materials, parts and assemblies shall be in accordance with applicable drawings and specifications.

3.2 Firing devices. - The firing devices shall comply with all requirements specified on drawings referenced on Drawing (dwg.) 8800924 and with all requirements specified in applicable specifications.

3.3 Embrittlement of springs. - Cap springs, actuator springs, armature springs, and bail springs shall not break when extended to the minimum (min) point of permanent distortion.

3.4 Air leaks. - Firing devices shall show no evidence of cracking or separation of the parts or failure of the seals.

3.5 Functioning. - Firing devices shall function as follows:

3.5.1 Ambient (70 degrees plus or minus 5 degrees F).

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a. Mechanism does not actuate when handle is depressed with bail in safety (vertical slot) position.

b. Mechanism actuates when handle is depressed with bail in functioning (angular slot) position.

c. Total energy generated shall measure not less than 16.3 milliwatt-seconds nor more than 27 milliwatt-seconds as dissipated through 1.76 ohms of a 3.25 ohm load. Total time to deliver this energy shall not exceed 24 milliseconds, nor be less than 10 milliseconds.

d. Actuating mechanism shall return to original cocked position and be capable of repeated use with a double pulse (positive and negative) for each actuation.

3.5.2 Cold temperature. - The firing devices shall function as specified in 3.5.1 a through d, after conditioning for a minimum of 24 hours at minus 40 degrees F. plus or minus 5 degrees F., with the exception that the minimum acceptable total energy generated shall be 12 milliwatt-seconds.

3.6 Workmanship.-All parts and assemblies shall be fabricated and finished in a thorough workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, crazes, unblended radii, porosity, warpage, burn marks, checks, chipped edges, blisters, excess flash, dirt, grease, rust, salt deposits, visible raised cement seams, solder splash, corrosion products, and other defects and foreign matter which would affect their serviceability. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat, legible and sharply defined.

3.7 First Article Inspection. - This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed

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requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product. - At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission. - Prior to the start of regular production the contractor shall submit a first article sample (see 6.1c) to a Government approved facility as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of component parts, sub-assemblies and assemblies which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing components, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract, and

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he shall submit a record of this inspection with the sample, including certificates of conformance of materials. A sample containing known defects shall not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit an additional sample or portion thereof of items for first article inspection. Until that phase of the first article sample consisting of parts, and firing devices is accepted, the contractor is in no way, authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.2.2 First Article Inspection Provisions

4.2.2.1 Component parts and Subassemblies

4.2.2.1.1 Submission. - Twenty-five (25) complete sets of parts (i.e. 25 each of every component part and subassembly, including at least three (3) items from each cavity used to produce plastic parts and be so identified as to permit tracing each item to the cavity in which it was molded).

4.2.2.1.2 Inspections to be performed. - The parts and sub-assemblies will be subjected by the Government to any or all of the applicable examinations and tests specified in 4.3.2, 4.3.3 and 4.3.5 of this specification and any or all of the requirements of the applicable drawings.

4.2.2.1.3 Rejection. - If any component or subassembly fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a component or subassembly to comply with any of the stated requirements.

4.2.2.2 Firing Device Assemblies

4.2.2.2.1 Submission. - Thirty (30) test sets completely packaged and sealed as required by the applicable drawings.

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4.2.2.2.2 Inspection to be performed. - The 30 firing device assemblies shall be subjected to the following tests in the sequence indicated in addition to the inspection of 4.3.2.18:

a. Air leak test (see 3.4) in accordance with the method specified in 4.4.2.

b. Functioning at ambient temperature (see 3.5.1) in accordance with the method specified in 4.4.3.

c. Functioning at cold temperature (see 3.5.2) in accordance with the method specified in 4.4.3.

4.2.2.2.3 Rejection. - If any firing device fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of the test set to comply with any of the stated requirements.

4.3 Inspection Provisions

4.3.1 Lot formation. - The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same material and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of firing devices shall contain parts from not more than one lot interfix number from one manufacturer.

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4.3.2 Examination. - The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Housing (see dwg. 8800936 covering a detail of dwg. 8837124).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None Defined		
Major:	AQL 1.0 percent		
101.	Inside diameter (cap cavity) ...	Gage	01001
102.	Diameter of middle cavity (actuator spring cavity)	Gage	01002
103.	Small outside diameter	Gage	01003
104.	Depth to bottom of middle cavity	Gage	01004
105.	Concentricity of cap cavity with small cavity	Gage	01005
106.	Concentricity of middle cavity with small cavity	Gage	01006
107.	Length from small end to shoulder of large outside diameter	Gage	01007
108.	Interior surface finish improper or incomplete	Visual	01008

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Categories	Defects	Method of Inspection	Code No.
Minor:	AQL 2.50 percent		
201.	Outside length of intermediate shoulder	Gage	01009
202.	Length from bottom of cap cavity to shoulder	Gage	01010
203.	Face or radius at bottom of cap cavity missing	Visual	01011
204.	Chamfer missing at cap end	Visual	01012
205.	Intermediate outside diameter	Gage	01013
206.	Surface finish improper or incomplete	Visual	01014
207.	Evidence of poor workmanship (see 3.6)	Visual	01015

4.3.2.2 Actuator (see dwg. 8800937 covering a detail of dwg. 8800926).

Categories	Defects	Method of Inspection	Code No.
Critical:	None Defined		
Major:	AQL 1.50 percent		
101.	Large outside diameter	Gage	02001
102.	Small outside diameter	Gage	02002
103.	Inside diameter	Gage	02003
104.	Diameter of hole	Gage	02004
105.	Concentricity of small outside diameter with inside diameter	Gage	02005
106.	Concentricity of large outside diameter with inside diameter	Gage	02006
107.	Length to slot	Gage	02007
108.	Length of slot	Gage	02008
109.	Width of slot	Gage	02009
110.	Parallelism of hole with large end	Gage	02010
111.	Length to hole	Gage	02011
112.	Surface finish improper or incomplete	Gage	02012

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Minor:	AQL 2.50 percent		
201.	Total length	Gage	02013
202.	Depth of inside diameter	Gage	02014
203.	Length of large outside diameter	Gage	02015
204.	Any radius missing	Visual	02016
205.	Evidence of poor workmanship (see 3.6)	Visual	02017

4.3.2.3 Plate, top (see dwg. 8800938 covering a detail of dwg. 8837124).

Categories	Defects	Method of Inspection	Code No.
Critical:	None Defined		
Major:	AQL 1.00 percent		
101.	Length to housing recess centerline	Gage	03001
102.	Length to coil recess	Gage	03002
103.	Length of coil recess	Gage	03003
104.	Small housing recess diameter	Gage	03004
105.	Height to top of coil recess ..	Gage	03005
106.	Angle of armature abuting faces	Gage	03006
107.	Total depth of housing recess .	Gage	03007
108.	Any screw hole missing	Visual	03008
Minor:	AQL 2.50 percent		
201.	Thickness at ends	Gage	03009
202.	Thickness at coil recess	Gage	03010
203.	Total length	Gage	03011
204.	Chamfer missing	Visual	03012
205.	Any radius missing	Visual	03013
206.	Surface finish improper or incomplete	Visual	03014
207.	Evidence of poor workmanship (see 3.6)	Visual	03015

4.3.2.4 Plate, bottom (see dwg. 8800939 covering a detail of dwg. 8800926).

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Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 Percent		
101.	Length to actuator recess centerline	Gage	04001
102.	Length to coil recess	Gage	04002
103.	Length of coil recess	Gage	04003
104.	Actuator recess diameter	Gage	04004
105.	Height to top of coil recess	Gage	04005
106.	Angle of armature abuting faces	Gage	04006
107.	Any screw hole missing	Visual	04007
Minor:	AQL 2.50 percent		
201.	Thickness at ends	Gage	04008
202.	Thickness at coil recess	Gage	04009
203.	Total length	Gage	04010
204.	Depth of counterbore	Gage	04011
205.	Surface finish improper or incomplete	Visual	04012
206.	Evidence of poor workmanship (see 3.6)	Visual	04013

4.3.2.5 Armature (see dwg. 8800940 covering a detail of dwg. 8800926).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 Percent		
101.	Total length	Gage	05001
102.	Width	Gage	05002
103.	Thickness	Gage	05003
104.	Height to hole centerline	Gage	05004
105.	Width of slot	Gage	05005
106.	Length to hole centerline	Gage	05006
107.	Diameter of holes	Gage	05007
108.	Chamfer missing at end of yoke	Visual	05008

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109.	Perpendicularity of hole centerline to armature length	Gage	05009
Minor:	AQL 1.00 percent		
201.	Surface finish improper or incomplete	Visual	05010
202.	Evidence of poor workmanship (see 3.6)	Visual	05011

4.3.2.6 Cap (see dwg. 8800941 covering a detail of dwg. 8800924).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Inside diameter	Gage	06001
102.	Outside diameter	Gage	06002
103.	Inside height	Gage	06003
104.	Outside height	Gage	06004
105.	Chamfer missing or incomplete.	Visual	06005
106.	Surface finish improper or incomplete	Visual	06006
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	06007

4.3.2.7 Pin, armature (see dwg. 8800943 covering a detail of dwg. 8800926).

Categories	Defects	Method of Inspection	Code No.
Critical:	None Defined		
Major:	AQL 0.65 percent		
101.	Length, head and groove	Gage	07001
102.	Shank diameter	Gage	07002
103.	Groove diameter	Gage	07003

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Categories	Defects	Method of Inspection	Code No.
Minor:	AQL 1.00 percent		
201.	Surface finish improper or incomplete	Visual	07004
202.	Evidence of poor workmanship (see 3.6)	Visual	07005

4.3.2.8 Spring, actuator (see dwg. 8800945 covering a detail of dwg. 8800926).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Outside diameter	Gage	08001
102.	Inside diameter	Gage	08002
103.	Solid height	Gage	08003
104.	Load at 0.33 inches	Gage	08004
105.	Load at 0.505 inches	Gage	08005
106.	Surface finish improper or incomplete	Visual	08006
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	08007

4.3.2.9 Spring, cap (see dwg. 8800948 covering a detail of dwg. 8800924).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Large outside diameter	Gage	09001
102.	Small inside diameter	Gage	09002
103.	Solid height	Gage	09003
104.	Load at 0.125 inches	Gage	09004
105.	Load at 0.391 inches	Gage	09005
106.	Ends not closed	Visual	09006
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	09007

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4.3.2.10 Support, armature spring (see dwg. 8800949 covering a detail of dwg. 8800926).

Categories	Defects	Method of Inspection	Code No.
Critical:	None Defined		
Major:	AQL 1.00 percent		
101.	Outside diameter	Gage	10001
102.	Length to hole	Gage	10002
103.	Hole not true centered	Gage	10003
104.	Diameter of hole	Gage	10004
105.	Radius missing	Visual	10005
106.	Hole not perpendicular to length	Gage	10006
107.	Surface finish improper or incorrect	Gage	10007
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	10008

4.3.2.11 Spring, armature (see dwg. 8837123 covering a detail of dwg. 8800926).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Outside diameter	Gage	11001
102.	Inside diameter	Gage	11002
103.	Solid height	Gage	11003
104.	Load at 0.535 inches	Gage	11004
105.	Load at 0.704 inches	Gage	11005
106.	Ends not closed and ground	Visual	11006
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	11007

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4.3.2.12 Prong (see dwg. 8845192 covering a detail of dwg. 8800925).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Diameter of contact end	Gage	12001
102.	Total length	Gage	12002
103.	Knurl missing or inadequate	Visual	12003
104.	Groove incorrect or inadequate	Visual	12004
Minor:	AQL 1.00 percent		
201.	Surface finish improper or incomplete	Visual	12005
202.	Evidence of poor workmanship (see 3.6)	Visual	12006

4.3.2.13 Top plate and housing assembly (see dwg. 8837124 covering a detail of dwg. 8800926).

Categories	Defects	Method of Inspectlion	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Height from plate face to end of housing	Gage	13001
102.	Perpendicularity of housing inside wall to plate face	Gage	13002
103.	Staking missing, incomplete or projecting excessively	Visual	13003
104.	Diameter of small cavity (Actuator cavity)	Gage	13004
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	13005

4.3.2.14 Coil assembly (see dwg. 8800934 covering a detail of dwg. 8800926).

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Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Soldering missing or inadequate	Visual	14001
102.	Any component missing loose, damaged or incorrectly assembled	Visual	14002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	14003

4.3.2.15 Armature assembly (see dwg. 8800926 covering a detail of dwg. 8800924).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Armature restrained from full parallel surface contact with either plate's abutment face	Visual	15001
102.	Actuator action incorrect when actuator is fully depressed by a cap (part #8800941) inserted in reverse direction and released (rough, binds, excessively loose, armature remains in "half hold" position	Visual	15002
103.	Armature restrained from full parallel surface contact with either plate's abutment face in its new position	Visual	15003
104.	Shims not aligned in contact with plate's abutment surfaces	Visual	15004
105.	Any component missing, loose broken or improperly assembled	Visual	15005
106.	Tape missing or peeling	Visual	15006
107.	Insulation tubing missing	Visual	15007

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Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	15008

4.3.2.16 Bail (see dwg. 8800933 covering a detail of dwg. 8800924).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Length	Gage	16001
102.	Height	Gage	16002
103.	Length of tip	Gage	16003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	16004

4.3.2.17 Firing device, prior to assembling right hand case (see dwg. 8800924).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Inadequate terminal solder connection (prior to application of adhesive and assembly into case half	Visual/ Manual	17001
102.	Excessive bare wire at terminal (prior to application of adhesive and assembly into case half	Visual	17002
103.	Retaining ring not properly seated	Visual	17003
104.	Cap spring incorrectly assembled	Visual	17004
105.	Any component missing	Visual	17005

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Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	17006

4.3.2.18 Assembly (see dwg. 8800924).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Bail spring missing or incorrectly assembled (bail does not snap back into vertical slot when in safe position)	Manual	18001
Major:	AQL 1.00 percent		
101.	Bail action incorrect (sticks, excessively loose) ...	Manual	18002
102.	Sealing compound missing, inadequate, raised at joints or excess not wiped off	Visual	18003
103.	Bail inadequately seated in vertical slot (excess flash or poorly formed slot)	Visual	18004
104.	Case cracked or split	Visual	18005
105.	Bail in functioning position (in angled slot)	Visual	18006
106.	Connector dust cap fits improperly	Visual/ Manual	18007
107.	Handle action incorrect (binds, excessively loose)	Visual	18008
Minor:	AQL 0.65 percent		
201.	Marking misleading or unidentifiable (see 3.6)	Visual	18009

4.3.2.19 Carton, prior to sealing (see 5.1.1).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Number of assemblies incorrect	Visual	19001
102.	Packing components missing	Visual	19002

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Minor: None defined

4.3.2.20 Carton, sealed (see 5.1.1).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Contents loose	Visual/ Manual	20001
102.	Carton damaged so that contents are exposed or liable to be exposed	Visual	20002
103.	Tape missing	Visual	20003
Minor:	None defined.		

4.3.2.21 Sealed case liner, prior to closing wooden box (see 5.2).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Waterproofness of case liner destroyed by rupture, puncture, separation or incomplete sealing	Visual	21001
Minor:	None defined		

4.3.2.22 Box, sealed wooden packing (see 5.2 and 5.3).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Strapping or hardware missing, broken or loose	Visual/ Manual	22001
102.	Box damaged to extent that contents are exposed or liable to become exposed	Visual	22002

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Minor:	AQL 1.50 percent		
201.	Contents loose	Visual/ Manual	22003
202.	Car seal missing, unsealed or improperly positioned	Visual	22004
203.	Marking missing, incorrect or unidentifiable	Visual	22005

4.3.3 Testing

4.3.3.1 Functioning (see 3.5)

4.3.3.1.1 Ambient temperature (see 3.5.1). - This test shall be performed 100 percent. Any firing device which fails to comply with the requirements listed in Table I shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.3 using test equipment in accordance with 4.3.4. (Non-Destructive)

TABLE I

DEFECT	CLASSIFICATION	CODE NO.
Mechanism actuates with bail in safe position (see 3.5.1a and 3.5.2)	Critical	18010
Mechanism does not actuate with bail in functioning position (see 3.5.1b and 3.5.2)	Major	18011
Total energy (see Note 1) (see 3.5.1c and 3.5.2)	Major	18012
Actuating mechanism does not return to original cocked position (see 3.5.1d and 3.5.2)	Major	18013

Note 1 - Duration of energy pulse to be tested in accordance with 4.3.3.1.3 at ambient temperature only.

4.3.3.1.2 Cold temperature (see 3.5.2). - Twenty (20) firing devices shall be selected from each lot for test. The lot shall be rejected if any assembly exhibits any defect listed in Table I. The test shall be performed as specified in 4.4.3 using test equipment in accordance with 4.3.4. (Non-Destructive)

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4.3.3.1.3 Duration of pulse (see 3.5.1c) - Major defect - Code No. 18014. - Twenty (20) firing devices shall be selected from each lot for test. The lot shall be rejected if any assembly fails to comply with the requirements. The test shall be performed as specified in 4.4.3 using test equipment in accordance with 4.3.4. (Non-Destructive)

4.3.3.2 Embrittlement of springs (see 3.3) - Major defect. Thirty (30) springs of each type shall be selected from each lot for this test. If one or more springs of one type fail to comply with the specified requirements, the lot shall be rejected. The test shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4. (Destructive test) Springs shall not be returned to the lot.

	Component	Code No.
a.	Cap spring	09008
b.	Actuator spring	08008
c.	Armature spring	11008
d.	Bail spring	23001
e.	Bail latch spring	24001

4.3.3.3 Air leaks (see 3.4) - Major defect - Code No. 18015. - This test shall be performed 100 percent. Any firing device which fails to comply with the specified requirements shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.2 using equipment and procedures in accordance with dwg. 9200408. (Non-Destructive)

4.3.3.4 Load test of male connector (see dwg. 8800925) - Major defect - Code No. 25005. - Fifty (50) connectors shall be selected from each lot for test. The lot shall be rejected if any connector fails to comply with the requirements. The test shall be performed as specified in 4.4.4 using test equipment in accordance with 4.3.4. (Non-Destructive)

4.3.4 Inspection Equipment. - The inspection equipment required to perform the examinations and tests prescribed in this section is identified under the "Method of Inspection" headings in 4.3.2 and 4.3.5 and test method 4.4 paragraphs herein (see 6.3).

4.3.5 Dimensional control of molded parts. - In place of the normal sampling and inspections associated with the Classification of Defects, and after a curing time for the parts has been approved (see Note 2) a sample of three (3) parts (as molded) from each cavity shall be fully inspected dimensionally to qualify a new or

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reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be inspected for at least the defects listed in paragraphs 4.3.5.1 through 4.3.5.5 after continuous production of each 5,000 parts or at the end of each week, whichever occurs first. Of the three (3) samples, one (1) sample shall be the last part produced. If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for the control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for inspection for each separate type of defect according to MIL-STD-105, using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. All cavities removed from production because of some fault, may after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the contracting officer to substantiate the request. Contractor design of gages and test equipment required to perform the inspections listed herein shall be forwarded to Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-QA-A-M for approval prior to manufacture of equipment. The following subparagraphs identify those items and minimum inspections subject to the requirements of this paragraph.

NOTE 2. - In establishing a curing time, dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30, 60, etc. mins.) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be sent to Picatinny Arsenal, Dover, New Jersey, 07801 ATTN: SMUPA-QA-A-M. If any new change in material, cycle time, a reworked cavity, or a new cavity is used, a new curing time shall be established and approved.

4.3.5.1 Connector assembly, male (see dwg. 8800925 covering a detail of dwg. 8800924).

Major Defect	Method of Inspection	Code No.
101. Depth to prong tips incorrect	Gage	25001

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102.	Distance between centerline of prongs	Gage	25002
103.	Cap does not fit securely	Manual	25003
104.	Presence of rubber or other foreign material on prongs	Visual	25004

Minor: None Defined

4.3.5.2 Case, right hand (see dwg. 8800947 covering a detail of dwg. 8800924).

Major defect		Method of Inspection	Code No.
101.	Depth to armature assembly support	Gage	26001
102.	Overall thickness	Gage	26002
103.	Height of armature assembly recess	Gage	26003
104.	Height from housing recess to pivot pin hole	Gage	26004
105.	Length to pivot pin hole	Gage	26005
106.	Bail groove missing	Visual	26006
Minor defect			
201.	Marking misleading or unidentifiable	Visual	26007
202.	Evidence of poor workmanship (see 3.6)	Visual	26008

4.3.5.3 Case, left hand (see dwg. 8837117 covering a detail of dwg. 8800924).

Major defect		Method of Inspection	Code No.
101.	Depth to armature assembly support	Gage	27001
102.	Overall thickness	Gage	27002
103.	Height of armature assembly recess	Gage	27003
104.	Height from housing recess to pivot pin hole	Gage	27004
105.	Length to pivot pin hole	Gage	27005

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Minor defect

201.	Evidence of poor workmanship (see 3.6)	Visual	27006

4.3.5.4 Handle (see dwg. 8800928 covering a detail of dwg. 8800924).

Major defect		Method of Inspection	Code No.
101.	Location of actuator tip	Gage	28001
102.	Thickness	Gage	28002
103.	Width of bail rest	Gage	28003
104.	Location of bail slot	Gage	28004
105.	Radius of bail slot	Gage	28005

Minor defect

201.	Evidence of poor workmanship (see 3.6)	Visual	28006

4.3.5.5 Boot (see dwg. 8800927 covering a detail of dwg. 8800924).

Major defect		Method of Inspection	Code No.
101.	Thickness of closed end	Gage	29001
102.	Outside diameter near flange	Gage	29002
103.	Wall thickness near closed end	Gage	29003

Minor defect

201.	Evidence of poor workmanship (see 3.6)	Visual	29004

4.4 Test methods and procedures

4.4.1 Embrittlement of springs. - The spring under test shall be stretched until the minimum point of permanent distortion is reached. If the spring breaks when extended to the point of distortion, it shall be classed defective. Springs tested for embrittlement shall not be returned to the lot.

4.4.1.1 Embrittlement of spring after testing. - Springs will be subjected to compression for a minimum of 24 hours at the assembled height before testing as specified in 4.4.1.

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4.4.2 Air Leaks. - Firing devices with dust cap off and bail in safe position shall be placed in a vacuum chamber, and a vacuum of 5.5 pounds per square inch minimum shall be drawn and held for at least one minute. Observe each boot and note each one which is not in expanded condition as compared to the normal condition. Release the vacuum and with bell cover removed observe each boot within 5 minutes noting each one which is in compressed condition as compared to the normal condition. Any item in which the boot is not in expanded condition while under vacuum, or is in compressed condition after release of the vacuum, does not comply with 3.4 and shall be rejected. All accepted items shall have the dust caps closed prior to packaging.

4.4.3 Functioning. - Functioning of the firing devices shall be determined as follows: Place bail in functioning position (in angular slot), depress handle fully to produce two pulses (positive and negative). Observe output to determine compliance with 3.5lb, c and d. (If the actuating mechanism fails to return to the original cocked postion during the cold functioning test, the rubber boot may be manually manipulated to cause return of the actuating mechanism.) Place bail in safe position (in vertical slot), and depress handle. Observe that bail prevents handle from being fully depressed and mechanism does not actuate (3.5.1a). The firing device shall be functioned within three (3) minutes after removal from the temperature conditioning chamber. Bail must remain in the safe position (in vertical slot) at completion of Functioning Test.

4.4.4 Load test of male connector. - With the connector held securely, the required force shall be applied at the soldering end of each prong separately in a smooth gradual manner so not to exceed its loading requirement as specified in the applicable drawings.

5. PREPARATION FOR DELIVERY

5.1 Packaging

5.1.1 Level A. - Fifty firing devices shall be packaged in a fiberboard box complete with half-slotted partitions assembled to form two layers of 25 cells in a 5 by 5 arrangement. A plain fiberboard pad shall be placed on the bottom of the box and on top of each layer of cells. The box shall conform to the requirements of PPP-B-636 for a Type CF, Class Domestic, Variety SW, Grade 200. The partitions and pads shall be constructed of the same fiberboard material as the box. Cell size

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shall be approximately 3 3/8 by 1 3/8 by 4 1/4 inches in depth, inside dimensions. Pads shall be equal to the inside length and width of the box less 1/16 inch. Box size shall be approximately 17 1/2 by 7 1/2 by 9 inches inside dimensions. Boxes shall be closed and a length of 2 inch side gummed, kraft tape, PPP-T-45, Type III, Grade B, shall be applied over the full length of the top and bottom center seam of the box with approximately 2 inch extension of tape onto the end panels of the box.

5.1.2 Level C. - Same as Level A.

5.1.3 Air Shipment. - When mode of transportation is by air, firing devices shall be positioned in cells - one row with connector end down and next row with connector end up alternating in this manner for all 5 rows. The top layer of devices shall be positioned into the cells in the same manner as the lower layer except the first row shall be with connector end up. In no instance shall like ends of the firing devices butt each other. Each box of firing devices shall be tested by a Gauss meter to insure that the firing devices have been properly arranged within the box and comply with the magnetic shielding requirements for air shipment as specified in MIL-S-4473. Boxes failing to meet the air shipping suitability shall be opened and the firing devices rotated and rearranged to change the magnetic field until the loaded box meets the air shipment requirements. Alternatively, firing devices may be shielded in accordance with regulations of MIL-S-4473 to meet air shipment requirements.

5.2 Packing

5.2.1 Level A. - Two boxes (100 firing devices) shall be packed in a nailed wooden box with case liner and fillers. The box shall meet the requirements of Specification PPP-B-621 for a Class 2, Style 2, 2-1/2 or 4 box for a Type 3 load. The case liner shall conform to MIL-L-10547, Type I, Grade B requirements. The fillers shall comply with Specification MIL-F-50449 requirements and shall be added to the sides and ends of the box between the unit boxes and the liner, and on top of the closed case liner as required for a tight pack. Box size shall be approximately 18 1/2 by 15 3/4 by 10 inches in depth, inside dimensions. The case liner shall fit the interior dimensions of the wooden box. Fillers shall be equal in size to the length and width of the surface to be covered less approximately 1/2 inch. The case liner shall be closed by heat sealing. Seals shall meet the heat-sealed seam test of MIL-P-116. Boxes shall be closed and strapped in accordance with requirements contained in the appendix of the box specification.

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5.2.2 Level C.-Unit containers shall serve as shipping containers except tape shall be applied over the full length of all top and bottom fiber box seams with approximately 2 inch extension of tape onto end pannels and around all corners of the box.

5.3 Marking.-Each unit box and exterior wooden box shall be marked in accordance with the requirements of MIL-STD-1169. In addition, each unit and exterior box shall have special caution labels applied in accordance with paragraph 5.5.15.8.1 thru 5.5.15.8.4 of MIL-STD-129 as applicable. Labels shall be waterproofed in accordance with MIL-STD-129 requirements.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for submission of first article samples.

6.2 Inspection Code Numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. They should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.3).

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

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b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor Designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and amy include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SMUPA-QA-A-M, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

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Preparing Activity:
Army-MU

Project Number: 1345-A-013

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